

CSCE 676 DATA MINING · MODULE 04 · LECTURE NOTES

Clustering at Scale

Groups you did not know about, and whether they exist.

MODULE OUTCOMES

- Explain the main families and what each assumes.
- Explain the algorithms that handle data exceeding memory.
- Explain why cluster evaluation is genuinely hard.
- Explain how to choose k , and why the question is sometimes wrong.
- Assess a clustering honestly.

1 Families

Family	What a cluster <i>is</i>	Where it fails
k-means and variants	Points near a centroid.	Clusters that are elongated, of unequal size, of unequal density, or non-convex.
Hierarchical (agglomerative)	A merge at some level of a dendrogram.	Large n — it is $O(n^2)$ in memory and $O(n^2 \log n)$ or worse in time.
Density-based (DBSCAN, HDBSCAN)	A dense region of any shape, with noise points left unassigned.	Densities that vary substantially across the data — one parameter cannot suit both regions.
Spectral	A well-connected component of a similarity graph.	The affinity matrix is $O(n^2)$, so it does not scale without approximation.
Mixture models (EM)	A component of a probability distribution.	The parametric form is wrong, and it will fit it anyway.

So 'are there clusters in this data' is not a well-posed question until you say what a cluster is — and different definitions give genuinely different and equally valid answers on the same data, which is not a defect of the methods but a property of the question. **This is the module's honest core**, and it is why §3's evaluation problem is hard rather than merely unsolved.

K-MEANS MINIMISES WITHIN-CLUSTER VARIANCE, AND THAT ASSUMPTION IS DOING A LOT OF WORK

Minimising the squared distance to centroids assumes clusters are roughly spherical, of similar size, and of similar density — none of which it checks, and all of which it requires for the objective to correspond to anything you would recognise as a cluster.

So it will happily partition a single elongated cluster into three, and merge two adjacent spherical ones into one — confidently, with a good objective value, and with no indication at all that anything went wrong.

And it always returns exactly k clusters, whether or not the data has any group structure whatsoever — which means a k -means result is not, by itself, evidence that clusters exist (§4's closing callout).

But it is fast, simple, well understood, and it scales — so it is the right first thing to run and the wrong only thing to run, which is the honest summary of its place in practice and is not a criticism.

2 At scale

MINI-BATCH K-MEANS

update the centroids from a random sample each iteration. Converges to a slightly worse optimum, far faster. Almost always the right default at scale.

BFR

assumes clusters are axis-aligned Gaussians, so a cluster can be summarised by (count, sum, sum of squares) per dimension. One pass: assign points to summaries, keep only the summaries. Memory independent of n .

CURE

represents each cluster by a sample of REPRESENTATIVE POINTS rather than by a centroid, so non-spherical clusters survive the summarisation. One pass, and it relaxes BFR's assumption at some cost in memory.

AND THE COMMON IDEA

summarise, discard the points, keep the summaries. Which works exactly when the summary is sufficient for the model's assumptions -- and fails silently when it is not.

Summarise and discard is the pattern, and the summary's sufficiency is precisely the model assumption being made. BFR and CURE differ exactly in what they assume a summary can capture: BFR's three moments are sufficient statistics for an axis-aligned Gaussian and lose everything else, whereas CURE's representative points retain shape at the cost of storing more. **Reading the summary as the assumption is the transferable idea**, and it recurs in Module 06's sketches.

3 Evaluation

INTERNAL MEASURES REWARD THE SHAPE THE ALGORITHM WAS OPTIMISING FOR

Silhouette, Davies–Bouldin, and the within-cluster sum of squares all measure compactness and separation — which is exactly what k-means optimises, so k-means scores well on them by construction rather than by being right.

And DBSCAN's correctly identified non-convex clusters score badly on the same measures while being, by any reasonable reading, the better answer — so the measure selects the algorithm rather than evaluating it.

So internal measures can compare parameter settings within one algorithm and cannot compare across families — which is the limit to respect, and is routinely violated in comparisons that report silhouette across methods.

And external measures need labels, which you do not have — if you did, you would be doing classification rather than clustering. So the honest answer is stability plus inspection plus a null comparison, which is what follows.

- **Stability under resampling.** Cluster two disjoint subsamples and check whether the same structure appears in both — a clustering that changes completely was fitting noise, and this is the single most informative check available.
- **Stability under perturbation**, including simply a different random seed — k-means producing a substantially different answer from a different initialisation is informative about whether the structure is there, and it costs one re-run.
- **A null comparison.** Cluster data with the same marginal distributions and no group structure, and compare your score against it — which is the only way to know whether your silhouette of 0.4 is good, since the scale has no absolute meaning.
- **Inspection.** Read fifty members of each cluster and ask whether the grouping means anything to somebody who knows the domain — which is slow, unglamorous, and finds things no statistic does.
- **And downstream utility**, if the clustering feeds a task with its own metric — which is the only fully objective measure available and should be preferred whenever it exists. The null comparison is the one people skip, and a silhouette of 0.4 means nothing until you know what structureless data of the same shape scores, which is frequently around 0.3.

4 Choosing k

- **The elbow method** — plot the objective against k and look for the bend. Frequently there is no bend at all, the curve being smooth, and the choice is then arbitrary while appearing principled.
- **Silhouette against k**, which is less arbitrary than the elbow and inherits §3's circularity — it will prefer the k that produces the most spherical partition.

- **The gap statistic**, which compares the objective against a null reference distribution at each k — **the most principled of these methods** and **the most computation**, since it requires clustering the null data repeatedly.
- **Stability against k** , which asks at which k the structure reproduces across subsamples — **and is the one worth doing**, because it answers the question you actually care about rather than a proxy for it.
- **Or: use a method that does not require k** — DBSCAN takes density parameters instead, and hierarchical clustering defers the choice to where you cut the dendrogram. **But swapping the parameter is not escaping the problem: DBSCAN's density parameters are as consequential as k and are considerably harder to interpret**, so the decision has moved rather than disappeared.

AND SOMETIMES THE DATA HAS NO CLUSTERS

Most real high-dimensional data is one connected mass with varying density rather than a set of well-separated groups — **and a clustering algorithm will partition it anyway**, returning exactly as many groups as you asked for.

So **'the data has three clusters'** is a claim that requires the null comparison (§3) — **and without it, the statement is a description of your algorithm's output rather than a finding about the data.**

And a partition can be genuinely useful without being real — **customer segments for operational purposes do not have to be natural kinds** to be worth having, provided you are clear about which claim you are making.

Which is the distinction to state explicitly: 'a useful partition' and 'discovered natural groups' are different claims with different evidential requirements, and **conflating them is this module's characteristic overclaim** (Module 13 §1).

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Mining of Massive Datasets, chapter 7 (free)	§2's scale algorithms — BFR and CURE, with the assumptions stated clearly.
Tan et al., chapters 7 and 8	§1 and §3 — and the cluster validity chapter is the best available treatment of the evaluation problem.
von Luxburg — A Tutorial on Spectral Clustering (free)	§1's fourth row, properly — and the connection to graph cuts, which Module 07 uses.
Ben-David, von Luxburg & Pál — A Sober Look at Clustering Stability (free)	§3's stability measure, examined critically — including when it does not work.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Cluster the same data with k-means, DBSCAN, and hierarchical**, and compare the partitions.
- 2 **Construct data k-means gets wrong** — elongated, unequal, and non-convex.
- 3 **Run k-means on uniform random data** and observe that it returns k clusters.
- 4 **Implement BFR's summaries** and confirm memory is independent of n .
- 5 **Show where BFR's assumption fails** and what CURE recovers.
- 6 **Compute silhouette for k-means and DBSCAN** on non-convex data, and explain the ranking.
- 7 **Run the null comparison** and report what structureless data scores.
- 8 **Test stability across two subsamples** and quantify the agreement.
- 9 **Plot elbow, silhouette, and stability against k** and compare what each suggests.
- 10 **Read fifty members of one cluster** and say whether the grouping means anything.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name five families and what each means by 'cluster'.
- 2 Why is 'are there clusters' not well posed?
- 3 What does k-means assume, and what does it do when the assumption fails?
- 4 Why is a k-means result not evidence that clusters exist?
- 5 What is the common idea behind the scale algorithms?
- 6 How do BFR and CURE differ?
- 7 Why are internal measures circular?
- 8 What can internal measures legitimately compare?
- 9 Give five things that actually work for evaluation.
- 10 State the distinction between a useful partition and natural groups.